Tissue Engeneering

Tissue engineering is a field of regenerative medicine that aims to repair or replace damaged tissues by combining cells, biomaterials and biological signals to help the body grow functional new tissue.

HOW DOES IT WORK?

Scientists place living cells onto a supportive structure called a scaffold. The cells multiply and organize themselves while signals guide their growth, gradually forming tissue that can function like the original.

THE 3 BUILDING BLOCKS

Cells — create new tissue

Scaffold — provides structure and support

Signals — guide cell growth and development

Together, they create the conditions needed for tissue regeneration.

WHERE CAN IT BE USED?

Tissue engineering is being explored for repairing skin, bone, cartilage, blood vessels and organs. Researchers are also developing lab-grown tissues that could eventually help treat injuries and reduce the need for donor tissue.

THE FUTURE OF REGENERATIVE MEDICINE

Imagine growing replacement tissue using a patient's own cells. Tissue engineering could make personalized treatments possible, potentially reducing rejection and creating new options for repairing damage that the body cannot heal on its own.